THE LONG EXACT HOMOLOGY SEQUENCE OF A SHORT EXACT SEQUENCE OF DIFFERENTIAL OBJECTS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file homology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument; the reduction is displayed explicitly as a diagram of complexes and the only step it delegates is the named same-chapter lemma carried on this card as a navigation stub.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The same-name check for this card scanned every staged Stacks chapter of this repository – all sources/stacks-*/pinned-original-source.tex files, the card's own chapter excluded – and the labels found are recorded in dedup.same_name_cross_chapter on the card.

Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the comment-type reference, slogan and history environments (native preamble lines 8-13), the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217, the actual last line of the file).

Allowed standard prerequisites (editorial).

- Differential objects and their homology. Same chapter, the definition of a differential object and of its homology. Supplies the objects of the short exact sequence, their differentials and the homology functors appearing in the displayed sequence.
- The long exact cohomology sequence of a short exact sequence of complexes. Same chapter, Lemma lemma-long-exact-sequence-cochain. Supplies the exact sequence attached to a short exact sequence of complexes, which the proof applies to the periodic complex built out of the differential objects.

Source reference. Original lemma 0117: sources/stacks-homology/pinned-original-source.tex, source lines 3407–3407. Navigation pointer invoked at proof line 5734; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let $\mathcal{A}$ be an abelian category. Let $0 \to (A, d) \to (B, d) \to (C, d) \to 0$ be a short exact sequence of differential objects. Then we get an exact homology sequence*

$$\ldots \to H(C, d) \to H(A, d) \to H(B, d) \to H(C, d) \to \ldots$$

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Proof. Apply Lemma 0117 to the short exact sequence of complexes

$$\begin{array}{ccccccccc} 0 & \to & A & \to & B & \to & C & \to & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \to & A & \to & B & \to & C & \to & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \to & A & \to & B & \to & C & \to & 0 \end{array}$$

where the vertical arrows are d. □